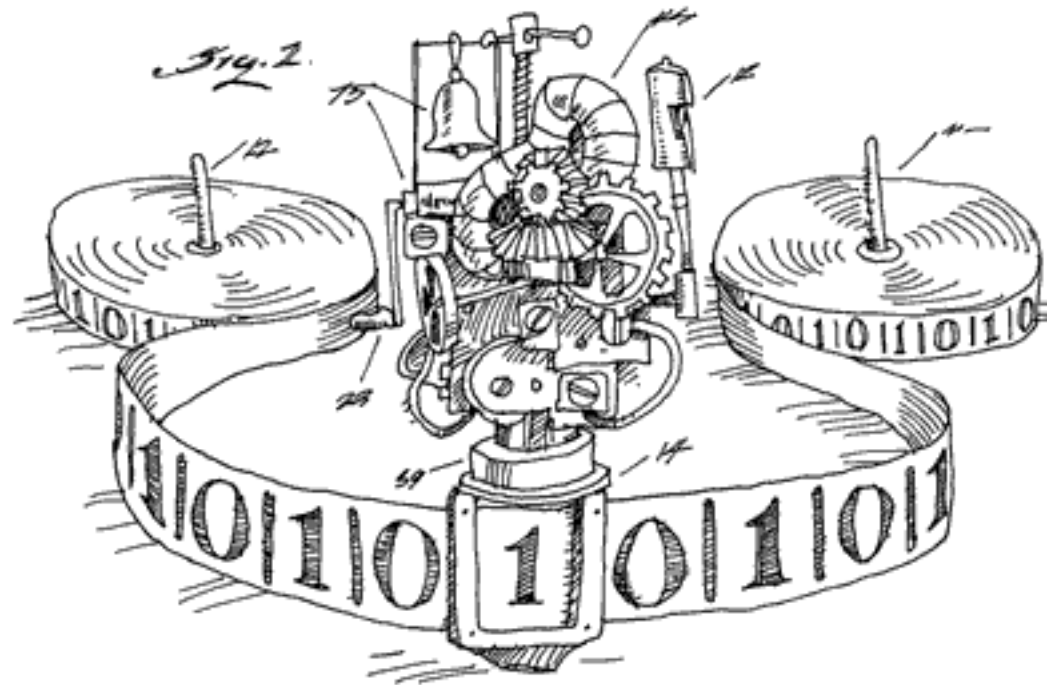


CSC4010: Introduction to Theoretical Computer Science



Tutorial 2

Question 1

1. In each of the following questions, choose either true or false.

(a) If $A \subseteq B$, then $|A| \leq |B|$.

☐ True

☐ False

(b) If $A \subsetneq B$, then $|A| < |B|$.

☐ True

☐ False

(c) Every subset of a countable set is countable.

☐ True

☐ False

(d) Every subset of a countably infinite set is countably infinite.

☐ True

☐ False

(e) Every language $L \subseteq \{0, 1\}^*$ is countable.

☐ True

☐ False

(f) The set of all languages $\mathcal{L} = \{L : L \subseteq \{0, 1\}^*\}$ is countable.

☐ True

☐ False

Question 2

2. Let $\mathcal{S} = \bigcup_{k \in \mathbb{N}} \mathbb{N}^k$ be the set of all finite sequences of non-negative integers, where $\mathbb{N} = \{0, 1, 2, \dots\}$. The case $k = 0$ corresponds to the empty sequence. Show that $\mathcal{S}$ is countable using one of the following methods. You may use the fact that $\{0, 1\}^*$ is countable.

1. Construct a one-one function

$$f : \mathcal{S} \longrightarrow \{0, 1\}^*.$$

Make sure to explain why your function is one-one.

2. Give an enumeration of $\mathcal{S}$ in which every element has a finite position.

Hint. For example, you may organize the sequences according to the value

$$k + n_1 + \dots + n_k$$

for a sequence $(n_1, \dots, n_k) \in \mathbb{N}^k$.

To prove B is countably infinite:

- a) Write $B = \cup_{i>0} L_i$, where $|L_i|$ is finite.
- b) Arrange elements of each L_i in a fixed order.
- c) The first element of L_i has index $\sum_{j=1}^{i-1} |L_j|$.
- e) We know position of $x \in L_i$.

Question 3

3. For an arbitrary set X , let $\mathcal{P}(X) = \{A : A \subseteq X\}$ denote the power set of X . Prove the following.

(a) Show that $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})|$.

(b) Prove that there is no bijection $f : \mathcal{P}(X) \longrightarrow X$. Conclude that $|\mathcal{P}(X)| > |X|$.

Hint: Prove by contradiction. Assume f is a bijection. Define $D_f \in \mathcal{P}(X)$ by $D_f = \{x \in X \mid x \notin f^{-1}(x)\}$.

